

reach greater depths with the help of new mechanical devices, such as dredgers for up to 18 m, operated by groups of bullocks and men, or steam hoists for deeper levels. Pneumatic caissons, a watertight enclosed structure used for construction in water, were used in places with challenging terrain, such as the Damodar Bridge on the Bengal–Nagpur Railway. It was built over a sandstone and hard shaley rock river bed, and opened for use in 1903. Technology later allowed workers to sink the piers even deeper, as with the Dufferin Bridge. Built over the River Ganges at Benaras, it opened in 1887 with piers that had been sunk to up to 49 m in some instances and was designed to accommodate both rail and road traffic.

Scouring of piers was a big problem for bridges built across rivers, so engineers used protective curtains or “aprons” made of piles of loose stones and cement ballast. They also started spacing out the piers. This became easier during the 1880s after iron, used for making girder spans, made way for steel.

Applying new learnings

In some places, especially where there were deep channels, engineers opted for the more complicated cantilever design, such as the Jubilee Bridge (now-decommissioned) that opened in 1885 over the River Hooghly in Calcutta. It was built using rivets and without nuts or bolts. The Lansdowne Bridge, which opened in 1889 and crossed the River Indus between Sukkur and Rohri (in present-day Pakistan) was a railway connection between Calcutta and Karachi. Designed by the British civil engineer Sir Alexander Rendel, the bridge had two anchored 52-m-high cantilever structures, with a suspended span of 61 m in the middle, making it an engineering marvel of its time.

Another significant development in construction was the practice of guiding rivers. This started during the 1870s with the building of groynes (low walls) for the Alexandra Bridge over the River Chenab (in present-day Pakistan). By the 1880s, bunds or Bell’s bunds – named after British engineer James Bell who devised the method – became popular. These involved building embankments with pitching stones, brought from quarries via rail or water. They protected against scouring